

Multiple Choice

1. **Answer:** A

Options: A) 0.119 wavelength; B) 0.130 wavelength; C) 0.112 *wavelength*²; D) 0.100 wavelength; E) 0.123 *wavelength*²

Note: Correct option: A - 0.119 wavelength

2. **Answer:** B

Options: A) 206; B) 193; C) 190; D) 212; E) 180 *Note: Correct option: B - 193*

2. **Answer:** D

Options: A) (4, 4, 0); B) (-2, 0, 0); C) (0, 0, 4); D) (4, 0, 0); E) (2, 0, 0)

Note: Correct option: D - (4, 0, 0)

3. **Answer:** A

Options: A) $[1 / (z - 1)]$; B) $[z / (1 + z)]$; C) $[z^2 / (z - 1)]$; D) $[1 / (2 - z)]$; E) $[(z - 1) / z]$

Note: Correct option: A - $[1 / (z - 1)]$

4. **Answer:** B

Options: A) 750 Hz.; B) 250 Hz.; C) 625 Hz.; D) 500 Hz.; E) 800 Hz.

Note: Correct option: B - 250 Hz.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $\text{Var}Z = \text{Var}X + \text{Var}Y - 2E[\{X - E(X)\}\{Y - E(Y)\}]$ '.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $U(t) = L_0 (1 + \text{sech}^2 t) \cos_{-0} (t + \tanh t) \sin$ '.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Nitrogen.'

Long Form Response

11. **Answer:** $L\{t^n\} = [n^2 / (s^{(n+2)})]$, for $s > 0$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $1/6$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key